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**B.Sc. (E.C.S.) (Semester - I) (NEP CBCS) Examination:
October/November – 2025
Numerical Methods (G06-0104)**

Day & Date: Thursday, 13-11-2025
Time: 12:00 PM To 01:30 PM

Max. Marks: 30

Instructions: 1) All questions are compulsory.
2) Figures to the right indicate full marks.

Q.1 Choose the Correct Alternative. (MCQ)

06

- While doing the Division of two numbers in normalized floating point the exponent's should be _____.
 - a) added
 - b) Subtracted
 - c) Divided
 - d) Multiplied
- If the data is equally spaced and interpolation is near the end of the data then _____ interpolation formula is used.
 - a) Newton's Forward Difference
 - b) Newton's Backward Difference
 - c) Newton's Divided Difference
 - d) Lagrange's
- The equation $x + 2y = 6$ is _____ equation.
 - a) linear
 - b) Tangent
 - c) transcendental
 - d) none of these
- A matrix having only one column is called as _____.
 - a) Row matrix
 - b) Column matrix
 - c) Scalar matrix
 - d) None of these
- If $F(a)>0$, $F(b)<0$ then root lies between interval _____.
 - a) [0,1]
 - b) [a, b]
 - c) [-1, -2]
 - d) None of these
- $\Delta K = \frac{\Delta y}{\Delta x}$ K is constant.
 - a) K
 - b) 0
 - c) 1
 - d) -1

Q.2 Answer the following questions. (Any three)**06**

- a) Prepare the forward difference table for following data:

x	2	4	6	8	10
f(x)	15	19	37	57	70

- b) Find the interval in which root lies between of the equation $x^5 + 5x + 1 = 0$.
- c) Define scalar matrix with example.
- d) Define Linear Equation with one example.
- e) Evaluate: $0.5247 E_6 \times 0.6930 E_{-10}$

Q.3 Answer the following questions. (Any two)**06**

- a) Find the root of equation $x^3 - 4x + 1 = 0$ by using bisection method, performing only three interactions.
- b) Evaluate:
- a) $1.4586E_{15} + 7.3254E_{15}$
- b) $1.215E_{-4} \div 4.5263E_{-4}$
- c) Define matrix with suitable example.

Q.4 Answer the following questions. (Any two)**06**

- a) Define absolute error, relative error & percentage error.
- b) If $f(1) = 5, f(2) = 11, f(5) = 29$, Then find $f(4)$ by Lagrange's interpolation formula.
- c) Find the root of equation $x^3 - 2x - 5 = 0$ by using Regula Falsi method, performing only two interactions.

Q.5 Answer the following question. (Any one)**06**

- a) Estimate $f(12)$ by Newton's Backward difference interpolation formula using following data
- | | | | | |
|-------|----|----|----|----|
| x: | 10 | 15 | 20 | 25 |
| f(x): | 14 | 18 | 28 | 42 |
- b) Solve following system by Jacobi's Iterative method perform only two interactions.
- $$27x + 6y - z = 85$$
- $$6x + 15y + 2z = 72$$
- $$x + y + 54z = 110.$$